MENG 4205.002

LAB 10: TORSION

Marcos Esparza, Ngai Thang, Chase Rheinlander, Stanley Galindo
University of Texas Permian Basin
Midland, Texas

ABSTRACT

The purpose of this lab focuses on measuring the deformation of cylindrical rods due to torsional loads. Using the MTB torsion unit, three different cylindrical rods were torsionally loaded in increments AISI 304 stainless steel, 60/38 brass, and 2030 aluminum. The incremental angular deformation of the rods was then measured using a dial gauge. The results showed that in the elastic region, the torque and deformation relationship is linear, and the expected torsional results were observed for all three materials. Although the primary expected measurement was to characterize the deformation, secondary measurements regarding data collection supported the principles of mechanics of materials through shear strain, torsional stiffness and the torque-twist relationship.

Keywords: Torsion, shear strain, torsion modulus, polar moment of inertia, angular deformation, elastic behavior, torque-twist relationship, MTB torsion unit

Nomenclature

T	*Applied torsion moment (N·m)*
θ	*Angular deformation (rad)*
γ	*Shear Strain (-)*
G	*Shear modulus of rod material (Pa)*
L	*Length of rod between clamps (mm)*
R	*Rod's radius (mm)*
D	*Rod's diameter (mm)*
I_p	*Polar moment of inertia (m^4)*
Load	*Mass loaded by lever arm (kg)*
Deform	*Dial gauge reading (mm)*

1. INTRODUCTION

Torsion loading occurs when an element experiences twisting from a moment or applied torque on the element [1][2]. This mode of deformation results in shear stresses that occur as a linear function from the axis of the rod to the outer radius. Many machine elements experience torsion loading, including drive shafts, axles, couplings and machine tool spindles [2]. Therefore, knowing how a cylindrical rod responds torsion is fundamental in designing safe and effective mechanical systems.

Deformation from torsion was measured for three cylindrical rods made of AISI 304 stainless steel, 60/38 brass and 2030 aluminum. Using the MTB torsion unit, incremental loads were applied through a known lever arm, and the resulting deformation was recorded using a dial indicator positioned at a fixed distance from the rotation axis. Since the purpose of this lab was to determine the amount of deformation from torsional loading, applied torque was compared to angular deformation from each material. The patterns of the data were proportional in the elastic region, as anticipated, due to the linear torsional behavior of cylindrical rods. In addition, the variance between the three materials occurred as anticipated due to their relative stiffness. Aluminum had the most deformity, followed by brass which deformed moderately, and lastly stainless-steel deforming to a minimum level per classical torsion theory [3].

2. THEORY

For a cylindrical rod subjected to torsion:

2.1 Angular Deformation and Shear Strain

The torsion angle θ relates to shear strain γ by:

$$R\theta \approx \gamma L \tag{1}$$

2.2 Hooke's Law for Torsion

$$\tau_{max} = G\gamma \tag{2}$$

Where τ_{max} is the maximum shear stress at the outer radius.

2.3 Torque-Stress Relationship

$$\tau_{max} = \frac{TR}{I_p} \tag{3}$$

For a circular rod:

$$I_p = \frac{\pi D^4}{32} \quad (4)$$

2.4 Combined Results

Equating the expressions gives:

$$G\gamma = \frac{TR}{I_p} \quad (5)$$

Substituting (1) and (5) gives:

$$G = \frac{LT}{I_p \theta} \quad (6)$$

This forms the basis for experimentally determining the torsion modulus.

2.5 Torsion Moment Calculation

Lastly from the manual [1]:

$$T = load\ (kg) \cdot 0.1m \cdot 9.8\frac{N}{kg} \quad (7)$$

2.6 Torsion Angle Calculation

$$\theta = \tan^{-1}\left(\frac{Deform(mm)}{60}\right) \quad (8)$$

3. METHODS

3.1 Preparation

The MTB Torsion Unit consists of a mobile clamp to set the testing length, a fixed clamp with a 0.1 m arm, a dynamometer for torque application, and a dial gauge 60 mm from the pivot point for deformation analysis as depicted in *Figure 2*. The three rods under assessment were all 8 mm in circumference to achieve consistency for accurate measurementation [3]. The mobile clamp was pushed in so that the exposed length of the rod was 350 mm. The half-moon clamps and Allen screws were then used to keep the rod fixed in place.

Then, with the mobile clamp and fixed clamp applied, the gauge was inserted and zeroed. Afterwards the dynamometer was applied to the level arm and torsional loading was applied in 0.25kg increments via the upper wheel. For each material, the following load applications were applied in sequence: 0.75 kg, 1.00 kg, 1.50 kg, and 2.00 kg, and the deformation read from the dial gauge was annotated for further analysis.

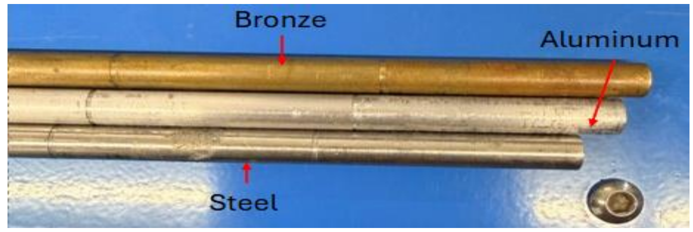

Figure 1 Bronze, Aluminum and Steel experimental rods from top to bottom respectively.

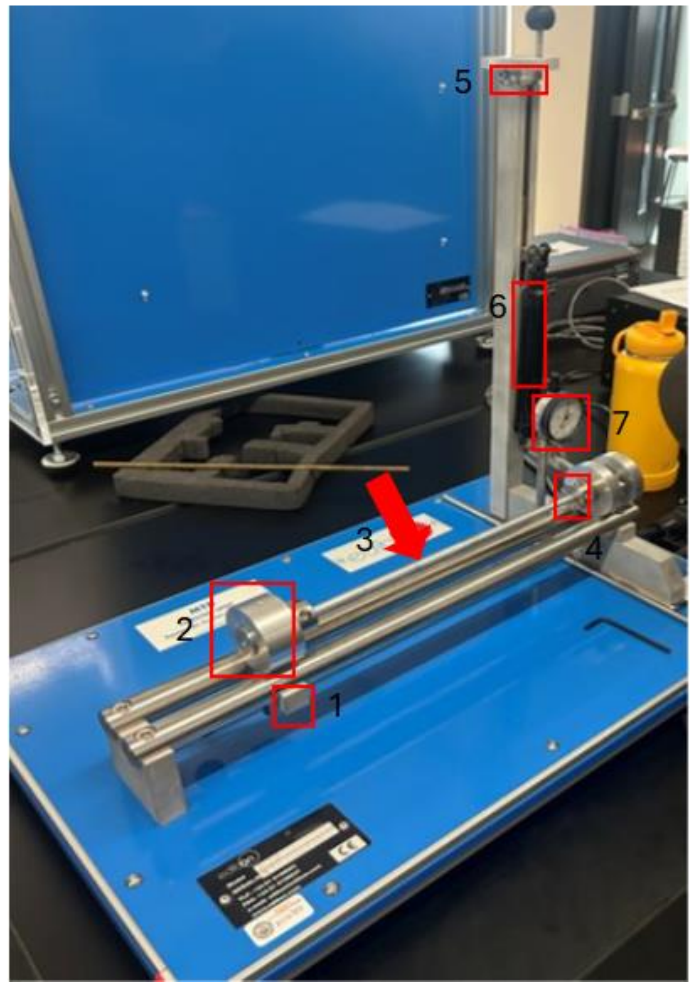

Figure 2 MTB Unit Apparatus. 1. Lower crank 2. Mobile clamp 3. Experimental rod placement 4. Half-moon clamps (one on each side of the rod 5. Upper wheel 6. Dynamometer 7. Dial gauge

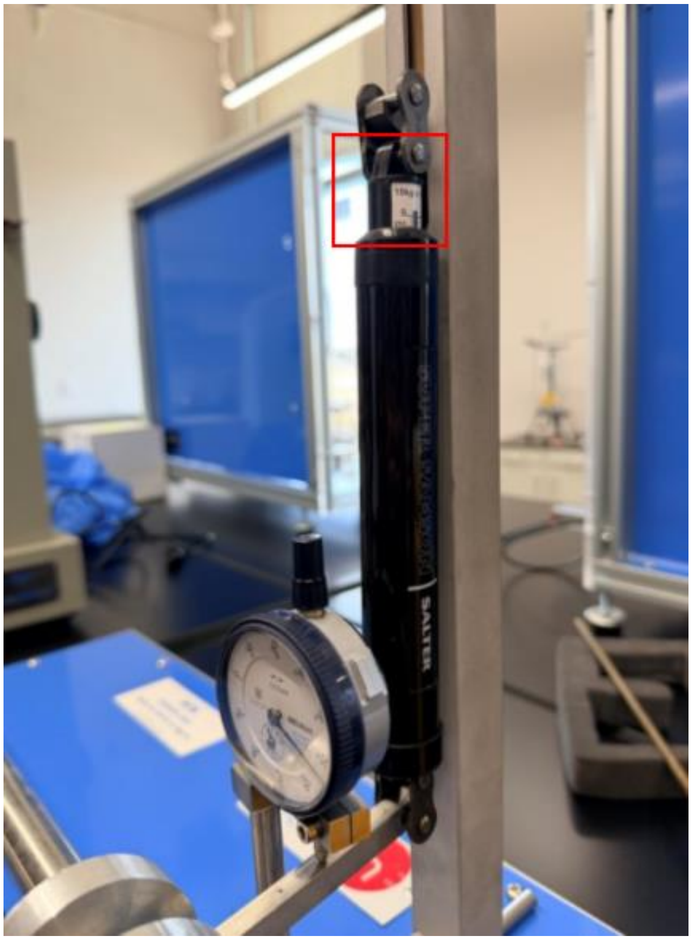

Figure 3 Applied charge readout

3.2 Data Reduction and Calculations

The measurement values recorded for each rod were the applied load (kg) along with the corresponding linear deformation measured by the dial gauge (mm) as shown in *Table 1*. To better understand how the applied torsion influenced the rod's deformation, the measured quantities were converted to engineering parameters relative to torsional loading. For instance, the applied load was converted to torque moment from the lever-arm measurements of the MTB unit, and applied deformation was converted to an equivalent relative angle of twist as shown in *Figure 4* and *5*. The dimension of the rods were then used to calculate the polar moment of inertia to better differentiate between the three materials of different properties. To further support and validate the deformation trends the shear modulus was also calculated (see *Table 3* and sample calculations).

4. EXPERIMENTS

4.1 Results

The applied loads resulting in linear deformations and computed angles of twist for brass, aluminum, and stainless steel are all recorded in *Table 2*. As the applied load increased from .75 kg to 2.00 kg in 0.25 increments, all three materials deformed proportionally. *Figure 4* demonstrates a linear trend for each material. The slope represents the torsional stiffness of each material and based on the graph it can be seen that stainless steel has the greats torsional stiffness, followed by brass and aluminum. *Figure 5* also represents a linear relationship, where the slope of each line represents the torsional slope which is subsequently used in *Table 3* to calculate the shear modulus G for each material. It can also be seen in *Table 3* that the experimental values were not too far off from the expected values which suggests that the measured deformation behavior followed the classical torsion theory.

Table 1 Measured deformation (mm) for brass, aluminum, and stainless steel under increasing loads

Load (kg)	Brass (mm)	Aluminum (mm)	Stainless Steel (mm)
0.75	0.27	0.35	0.19
1.00	0.55	0.68	0.35
1.25	0.85	1.09	0.48
1.50	1.08	1.47	0.61
0.75	1.26	1.65	0.82
2.00	1.40	1.80	0.93

Table 2 Converted deformation values in radians and corresponding torque for each material

Load (kg)	Torque (N·m)	Brass (rad)	Aluminum (rad)	Stainless Steel (rad)
0.75	0.74	0.004	0.006	0.003
1.00	0.98	0.009	0.011	0.006
1.25	1.23	0.014	0.018	0.008
1.50	1.47	0.018	0.024	0.010
1.75	1.72	0.021	0.027	0.014
2.00	1.96	0.023	0.030	0.015

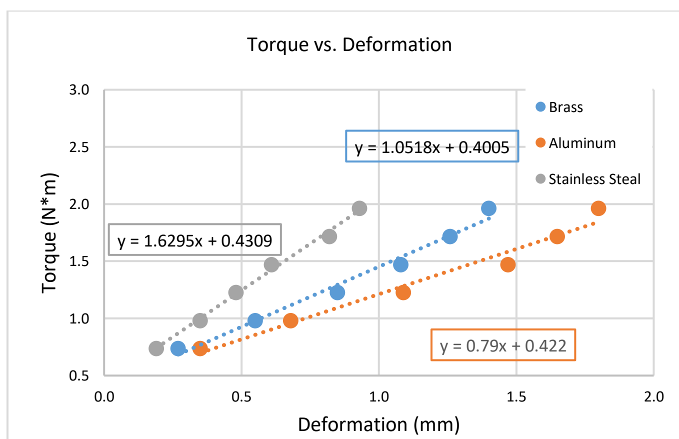

Figure 4 Torque increases linearly with deformation for all three materials

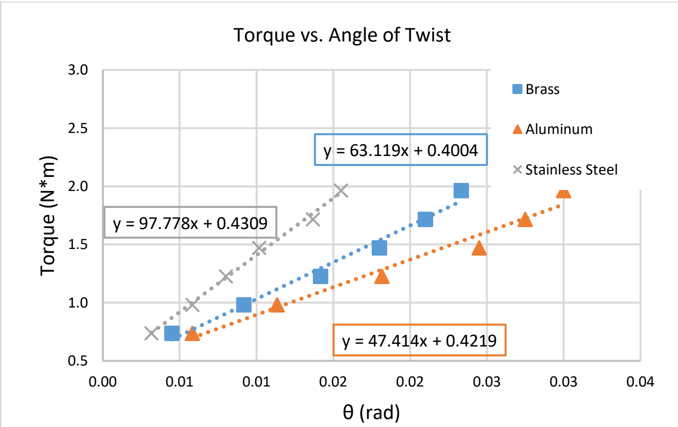

Figure 5 Torque varies linearly with angle of twist, allowing slope extraction for shear-modulus calculations

Table 3 Calculated shear modulus values for brass, aluminum, and stainless steel compared with ideal reference values

Material	$\frac{T}{\theta}(\frac{N\cdot m}{rad})$	I_p (m^4)	L (m)	G (GPa)	G_{Ideal} (GPa)
Brass	63.12	4E-10	0.35	55	35-40
Aluminum	47.41	4E-10	0.35	41	26-28
Stainless Steel	97.78	4E-10	0.35	85	74-77

4.2 Sample Calculations

Given:

Load = 2.0 kg

Deformation = 1.4 mm

Rod diameter = D = 8 mm

Rod length = L = 350 mm = 0.350 m

Torsion Moment

$$T = 2.0 \text{ kg} \times 0.1 \text{ m} \times 9.8 \frac{\text{N}}{kg}$$

$$T = 1.96 \text{ N·m}$$

Torsion Angle

$$\theta = \arctan\left(\frac{1.4}{60}\right)$$

$$\theta = \arctan(0.0075) \approx 0.023 \text{ rad}$$

Polar Moment of Inertia

Converting the rod's diameter in millimeters to meters:

$$D = 8 \text{ mm} = 0.008 \text{ m}$$

$$I_p = \frac{\pi D^4}{32}$$

$$I_p = \frac{\pi(0.008)^4}{32} = 4.02 \times 10^{-10} \text{ m}^4$$

Shear Modulus:

$$G = \frac{LT}{I_p\theta}$$

$$G = \frac{0.350(1.96)}{(4.02 \times 10^{-10})(0.023)}$$

$$G \approx 7.4 \times 10^{10} \text{ Pa}$$

This value (~ 74 GPa) is reasonable for materials such as brass or steel [2].

4.3 Discussion

The results of *Table 2* and the trends of *Figures 4* and *5* support the finding that all three materials had an increase in deformation relative to an increase in applied torque. This also corresponds with the expected outcome of a cylindrical rod within its elastic range. While deformation varied among the materials, the overall behavior was predictable. For instance, with the same amount of torque applied to each material, stainless steel deformed the least, followed by brass and aluminum.

In addition, the linearity of the graphs as shown in *Figures 4* and *5* also suggest that the rod samples twisted exactly as classical torsion theory indicates [4]. The slopes of the lines for *Figure 5* represent the torsional stiffness of the material which was used in *Table 3* to calculate the shear modulus G and this was grossly accurate to the expected G for each material. While calculating G was not necessarily the goal of this experiment, the findings therein were additional corroborating evidence that the deformation measurements captured the relative torsional rigidity of each material accurately.

The slight differences between the values of the shear modulus determined experimentally and theoretically were caused by a multiplicity of factors. The materials of the rods may not have been exactly what they were stated to be; nominal variations in alloy composition can cause appreciable differences in stiffness. Additionally, there could also have been some discrepancy when loading the rod. Furthermore, natural sensitivity limits of the dial gauge and human error in alignment and gauge readings also could reasonably have played a role. Despite these sources of uncertainty, the experimental results showed otherwise expected and appropriate behavior of deformation and stiffness rankings for all three materials aligning with the overarching theory.

5. Conclusion

The following experimental data obtained provided valuable insight in the deformation of cylindrical rods under torsional forces. Further is established the fact that in the elastic domain, angular deformation is directly proportional to the applied torque. As shown throughout the data, the deformation measurements for aluminum, brass, and stainless steel accurately reflected their relative stiffness, with aluminum twisting the most, stainless steel the least, and brass falling in between the two.

In addition, the calculation of the shear modulus further validated this claim as the material that had the highest rate of deformation resulted in the lowest shear modulus and vice versa.

Although slight variations in ideal values were detected, the general behavior of each material grossly abided by guiding classical torsion theory. The findings validate that torsional response of the tested rods was predictable, elastic, and in accordance with known mechanical properties of the rods.

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